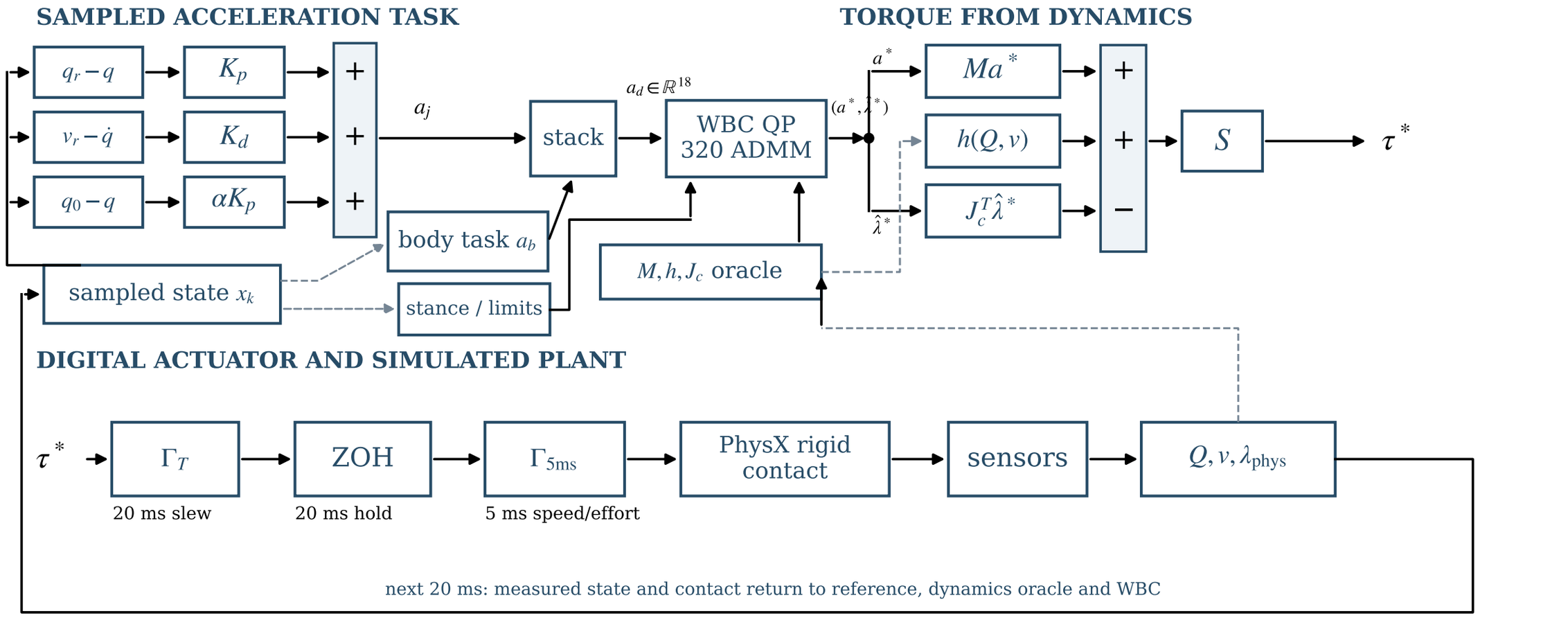
V16 | Actual WBC controller and physical loop

Whole-body QP job 577. Control update 20 ms; PhysX and emergency motor envelope 5 ms. All paths below are used by the runtime.



QP CONSTRAINTS AND LIMITS

$M_b a + h_b = J_b^T \hat{\lambda}$	$J_c a = a_c$	friction / normal force	motor torque bounds

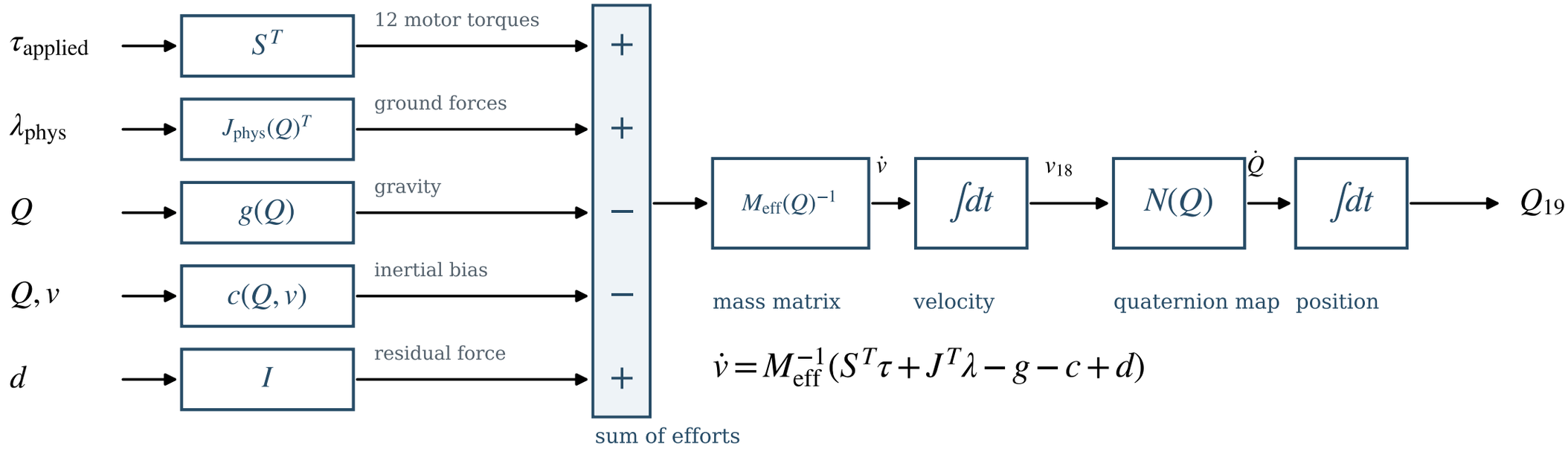
Gait/IK builds q_r; the dashed model path supplies h. QP and torque reconstruction share M, h, J. Predicted force differs from physical contact.

V16 | Physical plant built from small blocks

18-velocity-coordinate PhysX plant. Physical contact force comes from the contact solver, not the controller QP prediction.

EFFORT SOURCES

STATE EVOLUTION



$$\dot{v} = M_{\text{eff}}^{-1}(S^T\tau + J^T\lambda - g - c + d)$$

WHAT EACH BLOCK MEANS

$S^T\tau = [0_6; \tau_{12}]$ no base motor	$J^T\lambda$ foot force to generalized effort	M_{rigid} 54 masses, COMs, inertias	rotor armature 0.02 kg m^2 per joint

$$M_{\text{eff}} = M_{\text{rigid}} + S^T(0.02I_{12})S, \quad h = c + g, \quad \dot{Q} = N(Q)v$$

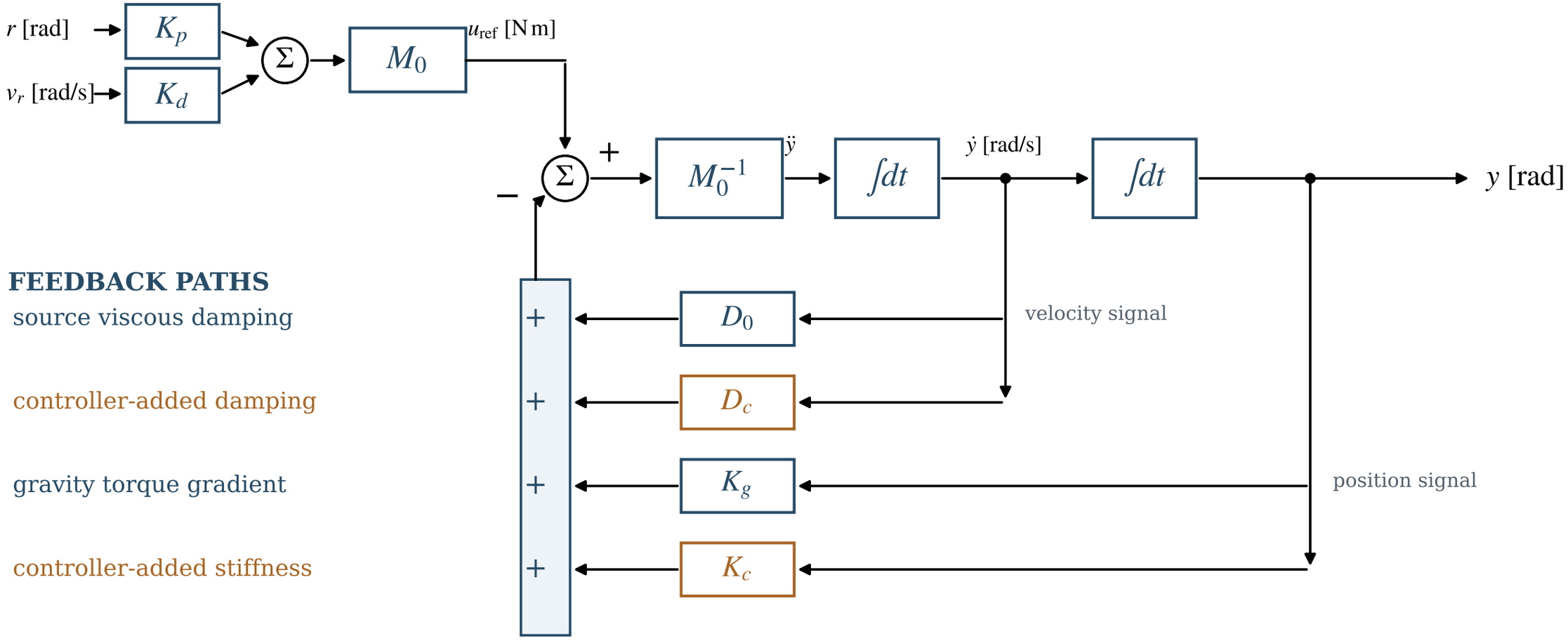
State-dependent M, J, g, c and N read the current Q, v; the outer state return is on page 1.

Actual USD joint damping is zero. d includes unmodeled joint friction. Contact forces can be distributed; J^T lambda is their generalized-force reduction.

V16 | Nominal 12-joint equivalence — analysis only

Fixed-base, no-contact source model; not the runtime WBC. Blue gains are physical, orange gains are ideal equivalent control terms.

REFERENCE TORQUE



COEFFICIENTS WITH PHYSICAL UNITS

$$D_c = M_0 K_d - D_0$$

$$K_c = M_0(1+\alpha)K_p - K_g$$

$$D_{\text{eff}} = D_0 + D_c$$

$$K_{\text{eff}} = K_g + K_c$$

D [N m s/rad], K [N m/rad]. Source D0 = 1.5 I; actual USD damping = 0. This nominal equivalence is not the runtime WBC on page 1.